The Price Is in the Protection

What 102 Sovereign Loan Files Show About Pricing, Security and Disclosure in Policy-Bank Lending

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Abstract. This paper assembles and codes the public record of 102 sovereign and sovereign-backed infrastructure loans across 34 borrower countries, with committed amounts of USD 240.1 billion, and asks three questions. What does a loan cost once every component of its price is counted? What does the lender take to protect itself, and what does that protection reveal? And what can an outside reader know about either, given what each side publishes? Three findings stand out. First, the headline interest rate is the smallest part of the price that can be measured: insurance premia running to 11.6 percent of facility value (a premium figure is held for only 6 of 102 deals), fees, trapped cash balances of up to USD 500 million on a single facility, and revenue capture all sit on top of it, and the premium is the rarest number in the archive (6 of 102, all six on Chinese facilities, none published by the insurer). Second, protection follows the lender, not the borrower: escrow or revenue capture is documented in 26 of 61 Chinese policy-bank deals and in none of 21 multilateral or DFI deals, the same sovereign receives a full revenue trap from one lender and an unsecured obligation from another, and in distress the cash traps that sat offshore kept working while no-comparability clauses did not survive a single systemic default and no asset pledge in the corpus is documented as enforced. Third, disclosure is the binding constraint: of 122 counterparties, 99 publish nothing at facility level, China Eximbank publishes no facility term for any of its 46 projects in the corpus, and the one window into a Chinese policy bank's distressed book, a Hong Kong regulator's template applied to China Development Bank's branch, shows a defaulted Venezuela position carried at exactly 25.00 percent allowance cover for six consecutive reporting dates before cover rose to 81.2 percent. A per-deal data-depth audit (102 deals by 13 dimensions) scopes every claim in the paper: depth follows documents, and documents follow law, leaks and distress, never deal size.

Keywords: sovereign debt; policy-bank lending; China Eximbank; China Development Bank; Sinosure; collateral; escrow; debt transparency; restructuring

JEL codes: F34, F65, H63, O19, G15.

Non-technical summary. When a government borrows from a foreign state lender to build a railway, a dam or a port, the number in the press release is an interest rate, often near 2 percent. This paper reads the actual paperwork for 102 such loans, worth USD 240 billion in total, and finds that the interest rate is close to the least informative number in the file. The borrower also pays an insurance premium that can exceed a tenth of the loan, posts cash into accounts the lender controls, and in many cases hands the lender first claim on the money the project earns, sometimes on taxes or oil shipments that have nothing to do with the project. What the lender takes in protection turns out to be a better guide to what the lender thinks of the loan than the rate it charges. The paper also documents a measurement problem. For most of these loans, an outsider cannot state the price at all: the largest lender in the sample publishes no terms for any of its 46 loans, the insurer names no claim it has ever paid, and for 96 of the 102 loans the insurance premium, usually the largest cost after interest, appears in no public document. Where the record does open, it opens for specific reasons: a parliament that must ratify loans, an auditor who publishes, a minister who releases a contract, a

loan that goes wrong. The paper measures that pattern deal by deal and reports which of its own conclusions the record can and cannot support.

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1 Introduction

A sovereign loan has two prices. The first is the interest rate, the number that governments quote and that datasets record. The second is everything else the borrower gives up: an insurance premium paid to the lender's own state insurer, fees, cash locked into accounts the borrower cannot touch, first claim on project revenue or on an unrelated tax or commodity flow, and clauses that govern what happens when the loan goes wrong. For most large policy-bank loans of the last two decades, the public record contains the first price and fragments of the second. This paper assembles those fragments for 102 loans, codes them on a common grid, and reports what the assembled record supports.

The sample is the three largest projects per borrower country across 34 countries in a disclosure census (Section 3), with committed amounts of USD 240.1 billion (Observed, B). The lenders are China Eximbank and China Development Bank on 61 deals, multilaterals and development finance institutions on 21, Russian state lenders on 3, and other bilateral or commercial lenders on 17. Every

deal carries a dossier: the contract text where one exists (13 deals), an official terms document (51), a partial contract (10), or reported terms only (28). Every deal is coded on 13 data dimensions, from the interest rate build to the governing law, and the coding distinguishes what a document states from what reporting asserts. The corpus also includes a counterparty archive of 438 banked documents covering 122 named parties, and a regulatory extraction from the Hong Kong disclosure statements of six Chinese banking institutions (340 position rows).

Four results organise the paper.

First, the headline rate is never the full price. Where an insurance premium is held at all, it runs from 5.26 to 11.6 percent of facility value (Observed, A), a one-off charge of the same order as several years of interest at the headline rate. In the best-documented case, Kenya's Standard Gauge Railway Phase 1, the borrower paid a Sinosure premium of USD 211.5 million and borrowed 85 percent of that premium inside the loan itself, so the borrower pays interest for up to 15 years on the cost of insuring the lender against the borrower. Trapped cash is a second hidden price: the same facility requires minimum balances of USD 250 million in each of two accounts during repayment, roughly USD 500 million of the borrower's cash held against a loan of USD 3.60 billion on the signed agreements' faces (USD 3.23 billion in National Treasury reporting). The premium is also the least disclosed component of price in the corpus: a premium amount or percentage is held for 6 of 102 deals, against 74 for the rate build and 87 for tenor.

Second, protection is a lender characteristic, not a borrower characteristic. Escrow or revenue capture is coded for 26 of 61 Chinese policy-bank deals and for none of 21 multilateral or DFI deals. Nepal borrowed for Pokhara airport from Eximbank with an escrow account attached and for the Tanahu hydropower plant from ADB, JICA and EIB with no capture at all, in the same credit, in the same period. Pakistan took no revenue security at all on a USD 2.89 billion Eximbank motorway credit. The variation runs by lender model and by the borrower's alternatives, and Section 6 documents it deal by deal across a 27-category taxonomy of mechanisms.

Third, the protection is informative precisely because the price is not. The insurer in this system behaves as an enforcement arm: Sinosure refused cover for Chad's Djermaya refinery as too risky, paid claims in Ethiopia (about USD 200 million (Observed, B)) and at Kafue Gorge Lower (USD 57.23 million, paid to the contractor that filed the claim), and in 2022 refused all new Pakistan cover until a revenue account promised in 2014 was funded, which produced a PKR 50 billion escrow five to seven years after the loans it protects had closed. In restructuring, the corpus shows a clean split: offshore cash traps survived and kept collecting, no-comparability clauses were overridden in the only systemic test they faced (Sri Lanka, 2024), and no asset pledge in the 102 deals is documented as enforced.

Fourth, none of this is visible from the lender side. Of 122 counterparties in the corpus, 99 publish nothing at facility level. Eximbank's 184-page 2025 annual report contains no overseas loan balance and no country figure for any of its 46 projects in the sample. Sinosure's 2025 report records claims costs up 24.6 percent and net profit down 86 percent, and names no claim. The single facility-level distress series for a Chinese policy bank anywhere in the public record exists because a host regulator's template forced it out of a branch: China Development Bank's Hong Kong branch carried its Venezuela advances at exactly 25.00 percent allowance cover, collateral recognised nil, for six consecutive reporting dates (December 2020 to June 2023) while the position was 100 percent overdue over one year at every date, before cover rose in two steps to 81.2 percent by December 2025 (Observed, A).

The paper's method contribution is the depth audit itself. Rather than treat missing terms as a footnote, we code every deal on the same 13 dimensions and let the audit scope the claims: a claim is

made only where the matrix holds the dimension that supports it, and the appendix prints the full matrix summary so any reader can check which deals carry which claim. The audit also produces a finding in its own right. Depth follows documents, and documents follow law, leaks and distress, never deal size: the THIN tier holds more committed money (USD 59.8 billion) than the MEDIUM tier (USD 54.9 billion), and the MINIMAL tier holds USD 13.6 billion, including a USD 4.3 billion dam of which only about USD 220 million is attributed to any lender in any document held.

Relation to the literature. The paper sits inside the empirical literature on Chinese overseas lending built by Gelpern et al. (2021) (henceforth the How China Lends project), which coded 100 Chinese loan contracts and established that confidentiality clauses, cross-default reach and Paris Club exclusion language are standard in that contract family, and Dreher et al. (2022) and Horn et al. (2021), who measure the scale and the cyclical pattern of China's overseas claims. Our corpus differs from How China Lends in construction: it is selected by borrower and project size rather than by contract availability, it extends beyond Chinese lenders to the multilateral, Russian and commercial deals the same borrowers signed, and it adds two layers that contract collections cannot supply, the lender-side counterparty archive and the Hong Kong regulatory series. The results agree with How China Lends on the contract template and extend it in one direction: the template's economic weight sits in its cash-control and insurance layers, whose prices and payouts are the least disclosed quantities in either corpus.

The rest of the paper proceeds as follows. Section 2 sets out the institutional background. Section 3 reports the disclosure census that defines the sample frame. Section 4 describes the corpus, the coding rules and the depth audit. Section 5 prices the loans component by component. Section 6 codes the protection. Section 7 reads the lender side, including the Hong Kong template and the Venezuela provisioning series. Section 8 follows the protections into restructuring. Section 9 classifies the corpus's anomalies. Section 10 states robustness checks and falsifiers, Section 11 lists limitations, and Section 12 concludes. Appendix A prints the depth matrix in summary, Appendix B lists all 102 deals, and Appendix C defines the sources.

2 Institutional background

Three institutional facts are needed before the data make sense. First, the lenders differ in kind. China Eximbank and China Development Bank lend under state mandates with funding advantages and, for Eximbank's buyer's credits, a standing practice of requiring export credit insurance from Sinosure, the state export credit insurer, with the premium charged to the borrower. Multilateral lenders (IDA, IBRD, ADB, EIB, AfDB, IsDB) lend under charters that require publication of signed amounts and, for JICA and JBIC, of term tables at signing. Russian state credits for the two nuclear plants in the sample are intergovernmental agreements reported only by the borrower side.

Second, the insurance is not a side contract. A Sinosure policy on a buyer's credit covers the lender, is priced as a percentage of the facility (principal plus interest is the insured base in several documented cases, so cover can exceed the loan's face), and gives the insurer subrogation rights and, in practice, a veto over new business with a borrower in arrears. Sections 6 and 8 document the insurer acting on that position.

Third, disclosure runs through borrower institutions, not lender goodwill. The channels that produced contract texts in this corpus are constitutional access rights litigated in court (Kenya), parliamentary ratification pipelines that publish agreements as law (Serbia, Montenegro, Cameroon, Ecuador), supreme audit institutions (Uganda, Zimbabwe, Kenya), gazette publication (Zimbabwe, Egypt), and one ministerial release after an election (Kenya, November 2022). No channel in the corpus is

Table 1: Disclosure census grades, 34 borrower countries (recipient publication only)

Grade	n	Countries
HIGH	5	Cameroon, Serbia, Ecuador, Philippines, Montenegro
MEDIUM	13	Kenya, Ghana, Mozambique, Sri Lanka, Uganda, DR Congo, Nigeria, Nepal, Indonesia, Egypt, Republic of Congo, Sierra Leone, Zambia
LOW	16	Angola, Ethiopia, Tanzania, Senegal, Chad, Guinea, Zimbabwe, Djibouti, Pakistan, Bangladesh, Laos, Cambodia, Maldives, Venezuela, Kazakhstan, Uzbekistan

Notes: Grades assess contract-level obtainability through the recipient state's own publication, per the census protocol in Appendix C. The universe of 102 deals covers the three largest projects per country.

Source: Kilambi Research disclosure census, October 2026.

a lender's voluntary facility-level publication, with two partial exceptions: the mandate publishers among the multilaterals, and the Hong Kong Monetary Authority's disclosure template applied to locally incorporated or registered branches, which is a host regulator's requirement, not a lender's choice.

3 The disclosure census

The sample frame is a census of recipient-side publication in 34 borrower countries: what the law in each country requires to be published about sovereign loan contracts, and what the state in fact publishes through its ratification process, gazette, debt register, audit institution and any contract releases. The census grades each country on recipient publication alone. Lender publication is excluded from the grading by design, because the census question is what a reader in the borrower's own public record can obtain.

The grades are HIGH for 5 countries (Cameroon, Serbia, Ecuador, Philippines, Montenegro), MEDIUM for 13 (Kenya, Ghana, Mozambique, Sri Lanka, Uganda, DR Congo, Nigeria, Nepal, Indonesia, Egypt, Republic of Congo, Sierra Leone, Zambia) and LOW for 16 (Angola, Ethiopia, Tanzania, Senegal, Chad, Guinea, Zimbabwe, Djibouti, Pakistan, Bangladesh, Laos, Cambodia, Maldives, Venezuela, Kazakhstan, Uzbekistan) (Observed, B). HIGH means full contract texts are obtainable for a substantial share of major foreign infrastructure loans through a domestic channel that exists now. Cameroon is the only whole-of-portfolio publisher in the census. MEDIUM means channels have produced full texts or clause-level disclosure for flagship deals, episodically. LOW means no realistic route to contract texts exists; terms surface only in aggregates, restructurings or leaks.

Three census results matter for everything that follows.

First, the paper law and the practice diverge in a consistent direction. Kenya's constitution (Article 35) and its Access to Information Act gave litigants an absolute right that the High Court upheld in May 2022; the SGR contracts still reached the public through a minister's release in November 2022, after an election, not through the court order. Egypt operates a gazette and ratification channel that the executive bypassed for a USD 3 billion World Bank loan and emptied of content for the USD 25 billion Dabaa loan. Zambia's constitutional debt-approval duty was read down by its Constitutional Court during the borrowing it was meant to govern and rebuilt by statute in 2022, after the fact. The census's working claim, stated there as a claim to be falsified in a second pass rather than as a

proven result, is that no access-to-information statute by itself has ever produced a sovereign loan contract in this universe. Contracts arrive through ratification pipelines, audit institutions, litigation plus politics, and distress.

Second, disclosure does not track indebtedness. The contract harvest assembled by the How China Lends project across more than 200 countries found 100 public Chinese loan contracts; within our 34-country universe the harvest is concentrated in Cameroon (23 contracts), Serbia (10) and Ecuador (8), with zero for Angola, Zambia, Mozambique, Ethiopia, Tanzania, Nigeria, Senegal, Egypt, Sri Lanka, Pakistan, Bangladesh, Laos, Cambodia, Indonesia, Nepal, Maldives, Kazakhstan and Uzbekistan, several of which hold larger Chinese loan stocks than the publishers (Gelpern et al., 2021). A deal-first research design that starts from available contracts therefore selects on the borrower's publication system. This paper's design starts from the census for the opposite reason: the three-largest-projects frame takes the deals as given and measures how much of each deal the record holds.

Third, the census names its dark zones rather than averaging over them. Angola (the largest Chinese borrowing stock in the sample frame, oil-collateralised, zero full contracts public), Laos (railway debt held inside a joint venture, among the highest Chinese-debt burdens relative to GDP in the world, documented only from creditor-side debt sustainability analyses), Pakistan (CPEC terms withheld even from Parliament), Venezuela (two contracts in the open against a stock an order of magnitude larger) and Ethiopia (executive-only disclosure, quantified publicly only through a Common Framework treatment) are stated as gaps. The depth audit in Section 4 shows the same pattern from inside the corpus: the LOW-census countries contribute the thinnest dossiers, but the correlation is not mechanical, because distress can force a LOW country's deal open (Chad's knowable loans are its rescheduled ones) and a HIGH country's channel can be discretionary per facility (Ecuador declassified its flagship CDB structures and declared other credit lines confidential in the same period).

4 Data and methods

4.1 Corpus construction

The deal corpus is the three largest projects by committed amount in each of the 34 census countries, 102 projects in total, where size is measured on documented close figures rather than announcements wherever the two differ (EACOP enters at USD 755 million documented closed, not the USD 1.0 billion announced; the correction is one of a class documented in Section 9). Committed amounts total USD 240.07 billion by the matrix sum (RECONSTRUCTED, B). One entry, Nigeria's Mambilla hydropower facility (USD 4.93 billion inside Eximbank's sole-lender total in the universe file), never closed; it is retained in the corpus and flagged at every use, because its presence in lender totals is itself a datum about how announced lending is counted.

Each project has a dossier built from the public record: contract texts, ratification and gazette publications, debt-office registers, audit reports, IMF programme documents, lender and counterparty publications, and reporting. Each dossier ends with a terms read: the deal's price components, security package, risk allocation and anomalies, written against the documents held. The counterparty archive banks 438 documents (345 PDF, 93 HTML) for 122 named parties, each with a SHA-256 register entry; 47 of the PDFs are scanned images, and all 47 were converted to text by OCR in a dedicated pass (RapidOCR at 150 dpi, every figure used cross-checked against the statement's own internal totals).

Sector composition is two broad groups. Grouping the universe file's unstandardised sector strings by

Table 2: Corpus summary statistics

Item	Value
Projects (deals)	102
Borrower countries	34
Committed amount, total (USD bn)	240.1
Committed amount, mean deal (USD bn)	2.35
Committed amount, largest deal (USD bn)	25.0 (El Dabaa, Egypt)
Deals by paper grade: full contract	13
partial contract	10
official terms only	51
reported terms only	28
Deals by lender class: Chinese policy-bank	61 (USD 134.7bn)
multilateral / DFI	21 (USD 36.4bn)
Russian state	3 (USD 38.7bn)
other bilateral / commercial	17 (USD 30.2bn)
Deals by region: Africa	57 (USD 105.4bn)
Asia-Pacific	33 (USD 93.7bn)
Latin America	6 (USD 36.0bn)
Europe	6 (USD 4.9bn)
Counterparty archive documents	438 (122 parties)
HKMA position rows extracted	340 (6 institutions)

Notes: Committed amounts are documented close figures where held, announced figures otherwise; Mambilla (USD 4.93bn) never closed and is included as coded, flagged in the text. Region and lender class are coded from the dossier lender string (Appendix C).
Source: DATA-DEPTH-MATRIX.csv and PROJECT-UNIVERSE.csv, computed by the desk.

desk keyword rules (a desk construction, applied uniformly), energy accounts for 53 deals and USD 134.4 billion, transport for 42 deals and USD 69.3 billion, with water (1), telecom (1), multi-sector (2) and unstated (3) making up the remainder. Mean data depth does not differ materially between the two large groups (6.1 dimensions held of 13 for energy, 6.2 for transport); escrow use does (40 percent of energy deals, 19 percent of transport deals), because the oil-backed and power deals carry revenue assignments that transport concessions more often replace with guarantees.

4.2 Coding rules

Each deal is coded on 13 dimensions: terms read completed; official terms document; full contract text; interest rate build; tenor and grace; fees; insurance premium (amount or percentage); escrow or revenue capture; security package; governing law and arbitration forum; confidentiality or no-comparability position; restructuring or renegotiation record; lender-side document. Each dimension is coded P (held from a document: contract text, ratification or gazette text, official terms publication, lender- or counterparty-issued document, or regulatory filing), R (circulating only as reported) or blank (not held, including every case where the dossier states expressly that the item appears in no paper obtained).

Three standing rules, applied uniformly, do much of the work. Documented absence counts as held for the position dimensions: Hambantota's absence of security is a finding because both credit texts were read in full, and a blank means the archive does not know, not that there is none. Premium requires an amount or percentage for the facility itself; "premium included, size unstated" does not count, and an ADB maturity premium, a rate component rather than insurance, is excluded wherever it appears. A lender-side document must be issued by a lender or counterparty, or be a regulator's disclosure of

Table 3: Corpus by sector group

Sector group	Deals	Committed (USD bn)	Mean dims held (of 13)	Escrow (%)
Energy	53	134.4	6.1	40
Transport	42	69.3	6.2	19
Water	1	0.3	6.0	0
Telecom	1	1.5	5.0	0
Multi-sector	2	9.0	7.5	50
Other / unstated	3	25.5	6.0	0

Notes: Sector strings in the universe file are unstandardised (41 distinct strings); groups are a desk construction by keyword rule, stated in the build record. Escrow held counts P and R codings. Source: DATA-DEPTH-MATRIX.csv joined to PROJECT-UNIVERSE.csv.

that lender; third-party research filed under a party's name does not count. The mechanical first pass was adjudicated deal by deal against the dossier evidence; 51 deals carry at least one override of the first pass, every override states its reason, and the override file is preserved with the build record. The recurring corrections are recorded in Appendix A because they define what the counts mean: interest rate swap premia and ADB maturity premia are not insurance premia; another project's premium quoted comparatively inside a dossier is not that deal's premium; concession tenors are not loan tenors; gap lists, which enumerate what is not known, were excluded as evidence throughout.

4.3 The depth audit

Depth tiers are assigned mechanically. DEEP requires an official terms document or full contract, the rate build present, and the security package documented. MEDIUM requires a document base with at least five dimensions present where the DEEP test fails, or at least seven dimensions without a document base. THIN is everything else with at least three dimensions; MINIMAL is two or fewer. The audit gives DEEP 48 deals (USD 111.8 billion), MEDIUM 23 (USD 54.9 billion), THIN 24 (USD 59.8 billion) and MINIMAL 7 (USD 13.6 billion) (Reconstructed, B). A refresh after the OCR pass and two coding completions moved exactly one deal, Busanga, from MINIMAL to THIN, when its partial contract was coded into the matrix (three documented positions, including its confidentiality clause); its price terms remain unheld, and the move is a documentation correction rather than new knowledge of the deal's economics. No other placement moved.

Figure 1 is the paper's measurement frame in one picture. The dimensions that describe a loan's existence (terms read, tenor, security package, official terms document, rate build) are held for 73 to 100 percent of deals. The dimensions that price its protection and govern its failure (premium 6, governing law 15, full text 13, lender-side document 30, escrow 30) are held for a minority. Sections 5 and 6 therefore rest on a documented subset, and the subset is not random: it is the ratification jurisdictions, the two contract-release cases, and the distressed deals. Every trend reported below is a statistic of that held subset with its n stated, and Section 10 tests which conclusions survive the selection.

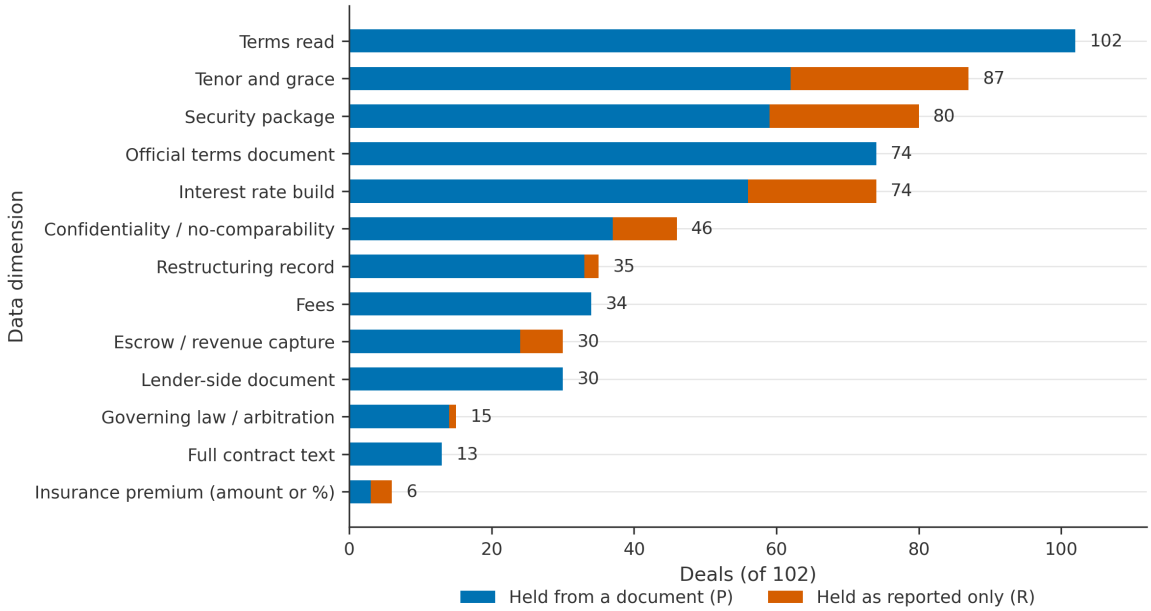


Figure 1: Coverage by data dimension, 102 deals

Source: DATA-DEPTH-MATRIX.csv. Note: P counts a dimension held from a document; R counts it held as reported only. The insurance premium, at 6 of 102, is the rarest priced quantity in the corpus.

5 The anatomy of price

A loan's price in this corpus has up to five components: the interest rate build, tenor and grace, fees, the insurance premium, and the opportunity cost of trapped cash and assigned revenue. The record holds them in sharply unequal measure. The rate build is held for 74 of 102 deals (56 from documents, 18 reported) and tenor and grace for 87; fees for 34, every one from a document and none from reporting alone; the premium for 6. The analysis follows the components in that order, and the pattern of what is missing is part of the result.

5.1 The rate build and tenor

Rate builds, where held, are conventional in form: a fixed concessional rate near 2 percent on government concessional loans, or a floating commercial build (a reference rate plus a margin) on buyer's credits, frequently inside the same project. Kenya SGR Phase 1 pairs a USD 1.6 billion concessional tranche with a commercial buyer's credit whose cap the released agreement states as USD 2,003.6 million, later revised to USD 1,903.4 million; National Treasury reporting puts the commercial tranche at USD 1.633 billion, a figure that reconciles with neither contract face. The two circulating rate figures the audit excluded by design mark the boundary of the data: Kafue Gorge Lower's 4.7455 percent and Nachtigal's modelled rates are World Bank Debt Reporting System weighted averages and an academic value-risk estimate respectively, not facility terms, and are coded not held. Anyone quoting a rate for Kafue is quoting an estimate of a rate. Tenor shows the same shape: held for 87 deals, but at Kafue the DRS average tenor is likewise coded not held, and three deals (Mambilla among them) have no tenor because no loan was ever agreed.

No harmonised rate series can be built from this corpus, and the paper does not build one. The matrix codes whether a rate build is held, not a standardised margin over a standardised base across

currencies and decades. What can be computed is the rate build's availability by vintage, and it falls: held for 86 percent of pre-2011 deals (n = 14), 83 percent of 2011 to 2014 deals (n = 29), 73 percent of 2015 to 2019 deals (n = 41) and 40 percent of 2020 to 2026 deals (n = 15) (RECONSTRUCTED, B). The recent fall is a disclosure lag as much as a disclosure choice: young deals have had no restructuring, audit cycle or ratification fight yet, and depth in this corpus accumulates through events (Section 4.3). The correct reading of the trend is that the public record of a loan's price decays as a share of new lending, not that new loans hide their rates more deliberately. Governing law follows the same vintage pattern more steeply (held for 29 percent pre-2011, 24 percent in 2011 to 2014, 7 percent in 2015 to 2019, 7 percent in 2020 to 2026), because forum clauses exist in the archive almost exactly where contract texts exist.

5.2 Fees: a document artefact

Fees are held for 34 of 102 deals, and the distribution by paper grade is exact: all 13 full-contract deals, 4 of 10 partial-contract deals, 17 of 51 official-terms deals, and 0 of 28 reported-only deals. No fee figure in the archive survives on reporting alone. The magnitudes, where held, are material to price: Ecuador's 2011 CDB facility states a management fee of 1.07 percent to the dollar; Kribi's full text states commitment and management fees of 0.3 percent each; Kenya SGR's released texts state management and commitment fees by tranche. Fees behave in the data as a pure artefact of document survival, which makes them a diagnostic: any cross-country comparison of loan pricing built from reported data alone will omit fees entirely for exactly the deals whose documents never surfaced, biasing the comparison toward the transparent borrowers. That bias is not correctable by weighting, because the missingness is the document, not a sampling probability.

5.3 The premium: the rarest number in the archive

Export credit insurance cover is coded into the security package of 22 deals. A premium amount or percentage survives for six: Kenya SGR Phase 1 (USD 211,543,142.85, P, released contract text), Kenya SGR Phase 2A (USD 161,875,409.70, 11.6 percent of facility, P), Hambantota (6 percent, P, in the credit texts), Kafue Gorge Lower (about USD 100 million, R, from a published project cost build-up), Karuma (about USD 65 million, about 10 percent of the buyer's credit face, R, from Ugandan parliamentary committee figures) and the AKK gas pipeline (7 percent, about USD 172 million, R, from the syndicate's Sinosure approval as coded in the dossier). Phase 1's basis needs stating exactly, because a wrong one circulates: the released agreement states the premium amount and no base for a rate. The 6.93 percent figure attached to this premium in AidData's coding of the agreement is stated there without a base and reconciles with no amount in the documents; against the caps the agreement does state, the amount is 10.56 percent of the buyer's credit's original facility cap and 11.11 percent of its revised cap. This paper therefore reports the Phase 1 premium as an amount, not a rate. A seventh figure circulates and is deliberately not counted: Sahiwal's 5.26 percent rests on an AidData facility record whose basis, principal or principal plus interest, is not stated, and the audit codes it not held; it sets the low end of the 5.26 to 11.6 percent range recorded in the collateral pass (OBSERVED, B). NEPRA's tariff file for the same plant allowed a Sinosure fee of USD 127.88 million, which is a regulator's allowance, not the premium paid.

Three properties of the premium matter more than its level. First, the borrower pays it, and in the Kenya case borrows it: 85 percent of the Phase 1 premium was financed inside the facility, so the premium itself accrues interest over the loan's life. Second, the insured base is principal plus interest in several documented cases, so cover exceeds the loan's face (Egypt NAC: USD 3.88 billion of cover

Table 4: Insurance premia held in the corpus (6 of 102 deals)

Deal	Premium	Basis	Grade
Kenya SGR Phase 1	USD 211.5m	Not stated in the agreement	P (contract text)
Kenya SGR Phase 2A	USD 161.9m (11.6%)	Facility	P (contract text)
Hambantota (Sri Lanka)	6%	Facility	P (credit texts)
Kafue Gorge Lower (Zambia)	about USD 100m	Facility	R (cost build-up)
Karuma (Uganda)	about USD 65m (about 10%)	Buyer's credit face	R (parliamentary)
AKK Pipeline (Nigeria)	about USD 172m (7%)	Facility	R (syndicate approval)

Notes: P = held from a document; R = held as reported only, per the coding rules of Section 4. Every one of the six attaches to a Chinese facility; no premium figure arrived from the insurer. For Phase 1 the agreement states the premium amount and no base for a rate; see the text.
Source: DATA-DEPTH.md premium note; COLLATERAL-PRACTICES.md cases 1, 4 and 10.

on USD 3.007 billion of principal; Atyrau: USD 1.7 billion on USD 1.13 billion; Maputo-Katembe: USD 1.096 billion face against USD 754 million of loans); commentary that treats cover above face as an error is misreading the product's arithmetic, and the informative figure is the premium rate, which is the figure borrowers disclose least. Third, every one of the six figures arrived through a borrower-side or secondary channel: released contracts, a parliamentary committee, a cost build-up, a regulator's tariff file. Sinosure's own publications price exactly two covers in the corpus's reach, Egypt NAC (USD 1,170 million, 2019 annual report, held from a read copy) and Cambodia's Pursat (over USD 200 million, 2025 annual report), and neither is a premium. For 17 of the 22 deals with documented cover, the cover exists in the record and its price does not. The Guinea RN1 dossier shows the shape of the hole: the agreement total exceeds the sum of its project shares by EUR 44.4 million, the dossier declines to assign the residual, and no paper states the premium.

5.4 Trapped cash as price

The fourth price component never appears in a rate table. Minimum balances and reserve accounts lock borrower cash for the loan's life. Kenya SGR Phase 1 requires USD 84 million minimum balances during grace, USD 250 million in each of the payment and revenue accounts during repayment, and USD 80 million in the tail: about USD 500 million trapped against a facility of USD 3,603.6 million on the signed agreements' faces (USD 3.233 billion in National Treasury reporting) at the repayment stage, before the Railway Development Levy (1.5 percent on all imports) is counted as a cost to the wider economy. Venezuela's family rule is a balance of at least 1.3 times the next repayment period's amounts; Ecuador's is 130 percent of the coming quarter's principal and interest, with required amounts rising from USD 50 million to USD 113 million; Ghana's CDB facility holds a debt service reserve at 1.5 times each repayment; Surgil's reserve accounts were funded at exactly their stated minima (USD 64.017 million debt service reserve, USD 56.071 million upstream reserve). These balances are priced in the contract and absent from every government's summary of its loan, and they are a real transfer: cash the treasury cannot use, held at the lender or its branch, in several cases offshore. A price comparison that omits them understates the cost of the escrowed loans relative to the unsecured ones by an amount that, for Kenya, is of the order of 14 percent of the facility's face in present cash terms.

6 The anatomy of protection

Security in this corpus means anything that improves the lender's position if the borrower stops paying: a trapped account, a pledged revenue stream, a guarantee, an insurance policy, a right to step in. The coding pass identified 27 categories of mechanism (26 mechanisms and one documented outcome, the contract that affirmatively takes nothing). The per-deal classification gives 59 deals documented (the package, or its documented absence, established by a contract text, an official terms source or an official financial record), 21 reported or inferred, and 22 unknown. This section works through the mechanism families, the two natural experiments the corpus contains (same borrower, different lenders; same project type, different account locations), and the case set a practitioner would stop at.

6.1 Cash control

The modal Chinese policy-bank package is control of cash before it reaches the borrower. Its strongest form is the offshore lender-controlled collection account: export proceeds are paid into an account at the lender itself or its offshore branch, and debt service is taken before the borrower sees the balance. Ecuador's 2011 CDB facility is the limiting case. Petroecuador sells oil to PetroChina; PetroChina pays CDB into a proceeds account at CDB in China; the committed delivery volume is fixed at 146,850,000 barrels notwithstanding any reduction in OPEC production quotas, so the obligation is denominated in physical oil and ranks in practice ahead of the country's OPEC commitments. The agreements acknowledge cross-agreement set-off while expressly denying that they create a security interest, a clean statement of the gap between legal form and substance that runs through the corpus.

Onshore escrow is the same instrument inside the borrower's jurisdiction, with lender withdrawal rights or an account bank that pays the lender directly. It is cheaper to operate and easier to sign, and it sits within reach of the borrower's own authorities, which Section 8 shows is the margin on which it fails. Between the two sit the reserve accounts: debt service reserves pre-funded at a stated cover (Ghana 1.5 times each repayment; Surgil's funded at exactly USD 64.017 million), maintenance reserves, and trapped minimum balances inside operating collection accounts (Venezuela's 1.3 times rule; Kenya's USD 250 million per account).

Revenue assignment without an account, a priority-of-application clause over project revenues, is the boilerplate form (Uganda's Karuma and Isimba: all revenues generated from the project shall be applied in priority to amounts due under the agreement). Dedicated levies extend the capture beyond the project: Kenya's Railway Development Levy charges 1.5 percent on all imports in the country to service a railway loan, and Guinea's RN1 road is secured on the bauxite mining tax revenues of three named Chinese mining companies, escrowed at ICBC Singapore, for a road that does not serve those mines in any way the documents describe. The account balance was EUR 76.35 million in both 2020 and 2021: not a flow passing through, but a trapped minimum sitting in Singapore while the road was built.

6.2 Commodity pledges, guarantees and insurance

Commodity offtake commits physical volumes, so a price fall increases the share of national output spoken for: Venezuela (200,000 barrels per day in 2010, rising to 300,000 from 2012, with a delivery top-up duty on PDVSA, which is not the borrower), Ghana's CDB facility (GNPC to sell 13,000 barrels per day of Jubilee crude to UNIPEC Asia for 15.5 years regardless of drawings), Congo (SNPC export proceeds into an Eximbank-controlled escrow) and Chad's Glencore prepayment, where the

creditor markets the cargoes itself and deducts debt service from the proceeds.

Guarantee structures are the multilateral system's main instrument, and they stack. Nachtigal, in Cameroon, layers an IBRD payment guarantee of up to EUR 86 million behind a Societe Generale letter of credit behind a state guarantee behind the offtaker's PPA payments, with a second IBRD guarantee of up to EUR 171 million wrapping the loans and MIGA cover issued at EUR 164.5 million. Bujagali, the corpus's contrast case, takes no offshore capture at all: an SPV borrower, a take-or-pay PPA, a government guarantee of the offtaker's payments, a utility-funded liquidity facility, direct agreements, and explicit contractual transfers of subsurface and hydrology risk to the state. The multilateral system's security is risk allocation written into contracts; the bilateral policy-bank system's security is cash held where the borrower cannot reach it. Both are coherent products. They are not the same product, and a borrower choosing between lenders is choosing between them.

Asset security is rarer than the public debate assumes. The pipeline itself is pledged on the Uzbek sections of the Central Asia-China gas pipeline and expressly not pledged on the Kazakh sections, where the package is account charges and a pledge of the ship-or-pay transportation agreement: same lenders, same sponsor, same asset class, opposite answers, which is the cleanest evidence in the corpus that collateral form is negotiated country by country rather than imposed from a template. Share pledges (Pakistan's two coal plants, Surgil, Chad's refinery), land and concession rights (Tema, Nacala, Dakar BRT), assignments of contracts and insurance proceeds, and step-in rights (Xayaburi, where the Thai offtaker EGAT may acquire the dam, with the Lao government agreeing not to intervene) complete the asset family. The counter-examples are findings, not gaps: Montenegro's Bar-Boljare Phase 1 is explicitly uncollateralised, Kenya SGR has no port mortgage, and Hambantota has no asset security in either of its two contract texts.

6.3 Contractual protections priced like collateral

Three clause families do economic work without any asset behind them. No-comparability clauses (Kenya SGR Art. 12.4; Kribi Art. 6.9; Coca Codo Art. 11.2, the strongest form; Hambantota in both texts; Kostolac B and the Hungarian-Serbian Railway) record the borrower's undertaking not to seek comparable terms in a restructuring; Section 8 records what happened when they met an actual default. Negative pledge and pari passu ratchets (Kribi and Novi Sad-Subotica Art. 6.1: any preference granted to anyone else is forthwith applicable to the agreement) import other creditors' bargains automatically. Consent rights reach further: Memve'ele Art. 12.2(d) gives the lender a consent right over the sovereign's other external borrowing. Opinion-based acceleration clauses convert political distance into a repayment event without any payment being missed: default defined by a material and adverse situation in the opinion of the lender (Hambantota GCL Art. 7.1(5), Montenegro Art. 7.1(5)), by harm to any entity of the lender's state including severed diplomatic relations (Ecuador 2011, cl. 18.13), or by changes in the lender's own government's policy (Serbia, Hambantota). A confidentiality or no-comparability position is documented for 46 of 102 deals; for 56 the archive cannot state even whether the terms are confidential.

Structural controls bind portfolios, not loans. Congo's single Eximbank escrow secures the whole 2006 framework portfolio, so no loan in it can be serviced or defaulted on separately; the trapped balances (USD 534.2 million in 2011, USD 513.8 million in 2018, USD 356.3 million in 2022) in several years exceeded half the RN1 loan's own balance, held offshore while the sovereign was in arrears to domestic suppliers. Lender-of-record structures separate the contracting party from the claim holder: Ghana's Sinohydro facility has the contractor as lender of record and China Construction Bank holding the receivable by assignment, with no contract between Ghana and CCB. Disbursement control enforces without clauses: proceeds paid straight to the contractor (Pakistan M5's pass-through accounts;

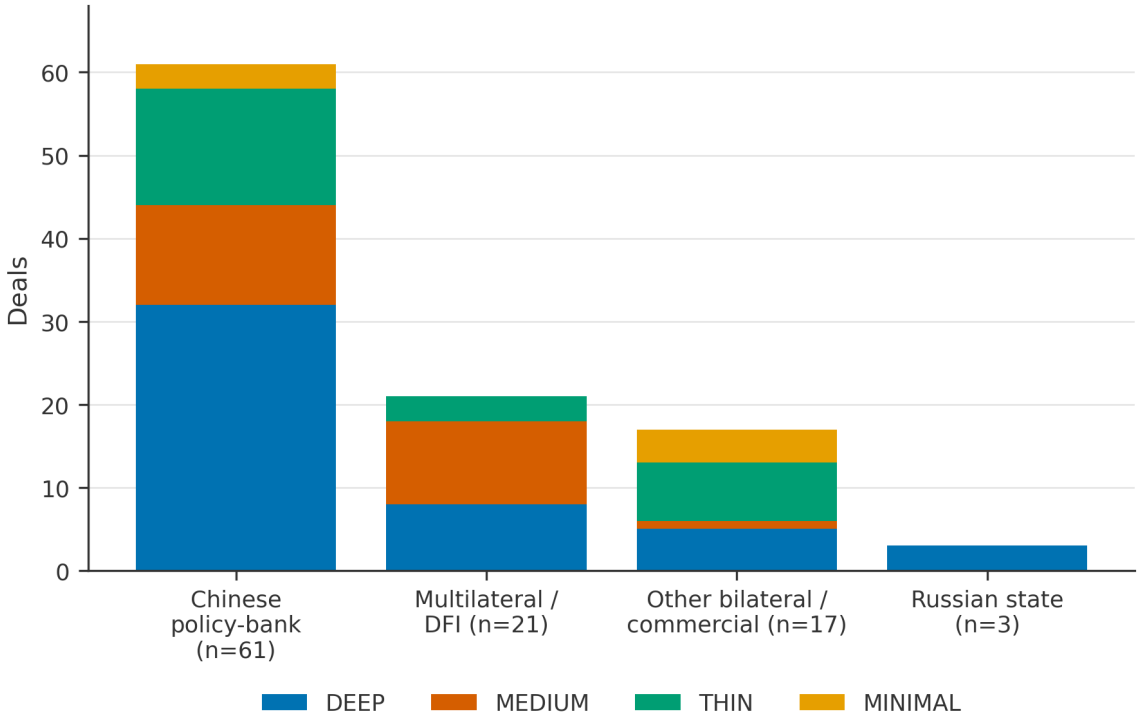


Figure 2: Depth tier by lender class (deals)
Source: DATA-DEPTH-MATRIX.csv. Note: Tiers are assigned mechanically (Section 4.2). Chinese policy-bank deals span the full range, because the class contains both the released-contract cases and the never-documented oil-backed structures; multilateral deals cluster at MEDIUM, deep on structure and security, thin on price.

Pokhara, where Eximbank pays the contractor directly and a Chinese project management company reviews every drawdown), and, in the corpus's strongest silent enforcement, USD 102 million of agreed liquidity withheld inside CDB accounts at Tonkolili in 2014 and 2015. The Rosatom credits carry a fuel-supply tie in place of any account or pledge: TVEL fuel for the plant's life, uranium at USD 550 per kg fixed to 2027 with decennial resets, and spent fuel repatriated, a dependency that functions as security in substance.

6.4 What the distribution shows

Escrow or revenue capture is held for 30 of 102 deals: 26 of 61 Chinese policy-bank deals, 0 of 21 multilateral or DFI deals, 1 of 3 Russian state deals and 3 of 17 other bilateral or commercial deals (Reconstructed, B). Of the 30, the account location is offshore for 11, onshore for 7, mixed for 2 (Karuma pairs Stanbic escrow accounts with a repayment reserve held at Eximbank; Beineu-Bozoi-Shymkent closed with an offshore charge and added an onshore pledge after a breach) and not stated for 10. The vintage pattern is the paper's template result: escrow held for 14 percent of pre-2011 deals (n = 14), 52 percent of 2011 to 2014 deals (n = 29), 29 percent of 2015 to 2019 deals (n = 41) and 7 percent of 2020 to 2026 deals (n = 15). The 2011 to 2014 band is where the cash-control template hardened: the corpus's dated clause families (Ecuador 2011, Kribi 2011, Kenya 2014) and the offshore account structures concentrate there, and later deals show thinner coded protection in the record. Two readings compete. The template may have loosened after 2014, or later deals may

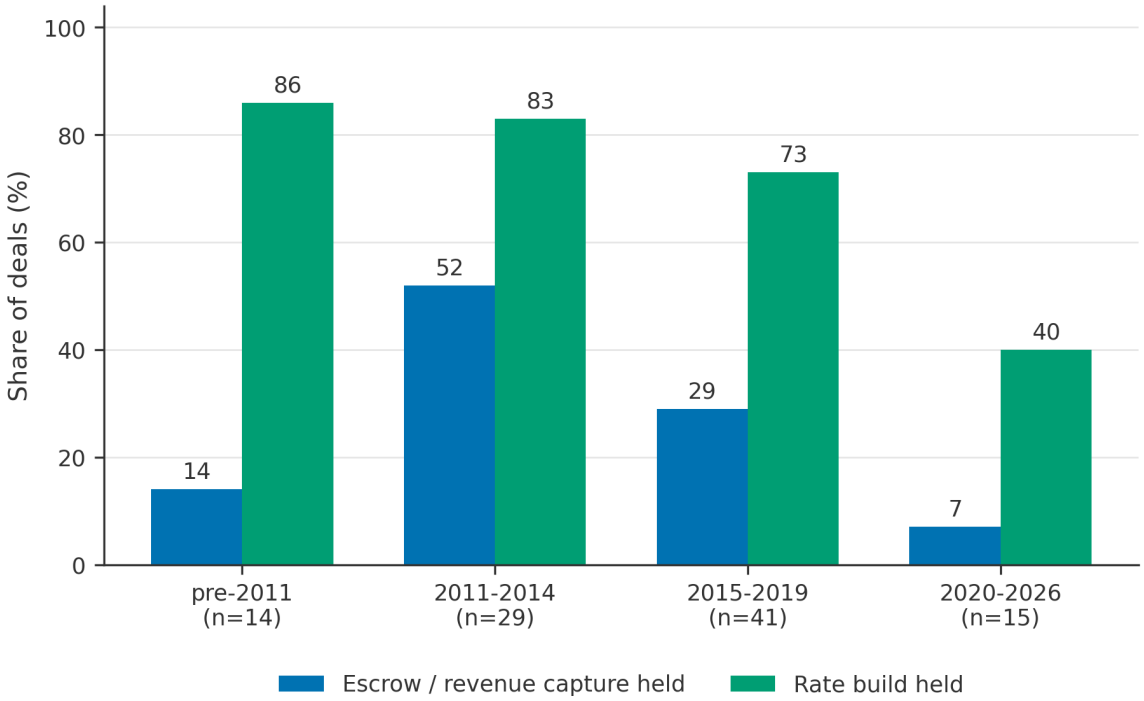


Figure 3: Escrow use and rate-build availability by signing vintage
Source: DATA-DEPTH-MATRIX.csv joined to PROJECT-UNIVERSE.csv signing years; 3 deals carry no signing year and are excluded. Note: Shares are of deals whose dimension is held (P or R). Escrow use peaks in the 2011 to 2014 band; rate-build availability declines monotonically across bands, a pattern Section 10 attributes in part to disclosure lag on young deals.

simply have had less time for audits, ratifications and restructurings to expose their packages. Section 10 separates the two as far as the data allow; the honest summary is that the peak is real in the held record and its cause is only partly identified.

The documented-absence category earns its place in the taxonomy. Nine deals carry full or partial texts that take nothing, or nothing beyond the sovereign obligation: Hambantota (both texts), Bar-Boljare Phase 1, Pakistan M5's buyer's credit, the Philippines South Commuter Railway (no escrow, no fare pledge, not even a negative pledge), Belgrade-Nis, Kostolac B, Tanahu, Olkaria and Matarbari. These are the deals that identify the variation. Security intensity does not track borrower poverty: it tracks the borrower's alternatives and the lender's model. Pakistan, with an IMF programme, competing creditors and a regulator publishing the financing chain's costs, signed an unsecured Eximbank motorway credit; the same lender took the full levy-and-escrow architecture in Kenya two years earlier. Nigeria's debt office states on the record that Chinese loans required no national-asset collateral, and its files are correspondingly thin, while the harshest packages sit where the borrower had no alternative funder and the asset earns foreign currency: Congo's oil framework, Venezuela's Joint Fund, Ecuador's CDB facilities, Guinea's tax escrow.

7 The China disclosure layer

7.1 The counterparty census

The borrower-side archive stands alone, and the counterparty pass measures how alone. For each of the 122 named parties in the corpus, the archive banks the party's own publications and codes what they prove at facility level. The result: 15 parties publish substantively (facility amounts or terms for archive projects), 2 are visible only through the Hong Kong template described below, 6 publish institutional aggregates only, and 99 publish nothing at facility level (Observed, B). The substantive 15 are overwhelmingly the multilaterals and export credit agencies whose charters require publication: IDA and IBRD (signed agreements in full), ADB, EIB, IFC, JBIC, JICA, MIGA, AfDB, IsDB, EBRD, GCF, DBSA, UKEF and EKN. JICA is the outlier of the entire corpus: full term tables (amount, rate, tenor, grace, tying status) at every signing, reconciling to the yen against borrower paper for Matarbari's eight tranches.

The three Chinese institutions at the centre of the lending publish, between them, no facility term in 438 banked documents. Eximbank (46 projects) publishes an aggregate only: its 2025 annual report runs 184 pages and contains an NPL ratio (0.95 percent), a fourth consecutive year of decline in the NPL balance, and no overseas loan balance, no geographic distribution and no country figure of any kind; its country-risk content is procedural. The English version of that report served on the bank's own site is a 5-page excerpt carrying printed pages 14 to 18, banked under the full report's title. CDB (20 projects) publishes group accounts with no overseas split of loans, allowances or NPLs; segment reporting is by business line. Sinosure (28 covers) publishes aggregates with two priced exceptions, both cover amounts rather than premia (Section 5.3). Bank of China (8 projects, USD 52.9 billion of commitments) publishes nothing at facility level; CCB (4 projects) publishes nothing at facility level, including for the Ghana receivables assignment where it is the effective cash financier.

The annual-report layer is accurate and uninformative at the same time, and the corpus establishes that this is a property of construction rather than of falsity. CDB's group NPL ratio halved over four years, 0.84 percent (2021) to 0.34 percent (2025), while its loan book grew 24 percent. China's National Audit Office found, in its FY2024 rectification report (p. 15), that two policy banks reclassified RMB 3.748 billion of loans as non-performing after audit: a loan-level read exists inside the system and is published as a violation class with the institutions' exposures never named. Aggregates do not contradict the facility record anywhere in the corpus. They are built so that the facility record cannot be inferred from them.

Sinosure's own series adds a trend the deal files cannot show. The medium and long-term book, the product covering this corpus, fell 74 percent from its 2018 level by 2021 (USD 23.55 billion to USD 6.17 billion insured) and recovered only to USD 8.25 billion in 2022, when new medium and long-term business was 127 projects totalling USD 9.30 billion. The insurer's centre of gravity moved to short-term trade cover. In 2025 claims cost rose 24.6 percent to RMB 14.98 billion and net profit fell 86 percent to RMB 166 million, and the report names no claim, by project, country or sector, in that year or in 2022's; of the 2022 claims total the report attributes USD 490 million to Belt and Road business and USD 890 million to short-term export credit, leaving the project book's claims unpublished in any year. The contrast inside the corpus is EKN, a smaller insurer, confirming in its own publication that it is paying indemnifications on Ethiopian transport cover after the 2023 debt relief. The information Sinosure withholds exists in the same genre at a peer institution. Its absence is a choice, not a constraint of the form.

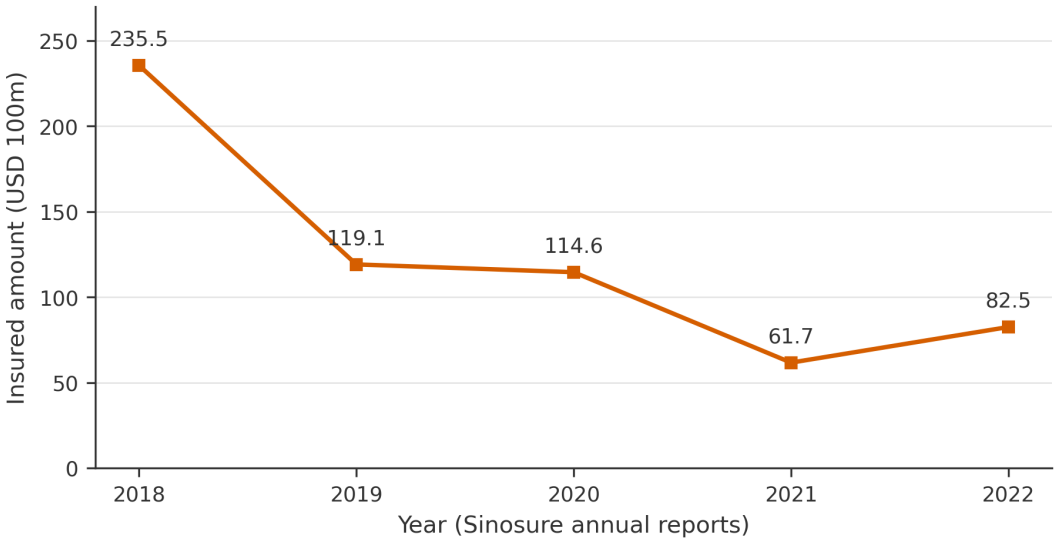


Figure 4: Sinosure medium and long-term export credit insurance, insured amounts, 2018 to 2022
Source: Sinosure Annual Report 2022, five-year product table (banked). Note: USD 100m units. The medium and long-term product is the cover that attaches to this paper's project loans; short-term export credit cover stood at USD 745.2 billion in 2022.

7.2 The Hong Kong template

The one mechanism in the corpus that forces facility-adjacent numbers out of Chinese institutions is a host regulator's disclosure template. The Hong Kong Monetary Authority requires branches and subsidiaries to publish financial disclosure statements with advances and impaired advances by geography. Applied to the Hong Kong grid's seven institutions, and extracted into 340 position rows for six of them, the template yields the corpus's only named-country impaired position in the Chinese system, a second named-country channel (ICBC (Asia)'s countercyclical buffer tables, which report risk-weighted exposures by jurisdiction and name Cambodia, Guinea, Indonesia, Kazakhstan, the Philippines, Bangladesh and Nepal), and a set of behavioural series that no statement narrates. Philippines RWA at ICBC (Asia) ran from HK$544 thousand (2018) to HK$524.5 million (2023) to HK$189 thousand (2025): a near-total exit with no announcement. CCB (Asia)'s residual "Others" geography held HK$44.7 billion of gross advances for one half-year (June 2019, against HK$16.1 billion a year earlier and HK$18.6 billion a year later) with impaired nil throughout. Bank of China (Hong Kong)'s "Others" exposures grew from HK$622 billion (December 2023) to HK$954 billion (December 2025) and cannot be decomposed from any document held; the bucket includes ASEAN subsidiaries, so the group's non-Hong Kong, non-mainland, non-US distress is permanently pooled and permanently unnamed.

Two behavioural signatures recur across institutions. Impaired balances freeze: Agricultural Bank of China Hong Kong Branch carried Hong Kong impaired advances between HK$87.8 million and HK$89.8 million for thirteen consecutive periods (2018 to 2024) and a mainland impaired balance of about HK$22.7 to 23.0 million for seven periods, before the mainland figure rose to HK$1,679 million at June 2024 and returned to nil by December 2025; its statements add that head office has provided no country-risk provision against the branch's exposures, the assessment being made centrally and published nowhere. ICBC Hong Kong Branch's entire disclosed problem book for a decade was one small Hong Kong loan, frozen at exactly HK$102.748 million for four consecutive reporting dates and

Table 5: CDB Hong Kong Branch: Venezuela (BANDES) impaired advances to banks

Period	Gross (HK$ m)	Allowances (HK$ m)	Cover (%)
Dec 2020	12,685.9	3,171.5	25.0
Jun 2021	12,831.6	3,207.9	25.0
Dec 2021	13,058.9	3,264.7	25.0
Jun 2022	12,484.9	3,121.2	25.0
Dec 2022	11,952.7	2,988.2	25.0
Jun 2023	11,580.5	2,895.1	25.0
Dec 2023	11,781.8	4,865.9	41.3
Jun 2024	11,698.5	8,483.7	72.5
Dec 2024	11,529.7	8,361.3	72.5
Jun 2025	11,707.8	n.p.	n.p.
Dec 2025	11,821.0	n.p.	81.2
Jun 2026	12,292.8	n.p.	81.2

Notes: n.p. = allowance amount not printed in the extraction table for that period; cover at Dec 2025 and Jun 2026 is as reported in the full-corpus read. At every period the position is 100 percent overdue over one year, collateral recognised is nil, and no statement records a repayment, write-off or restructuring. Gross at Dec 2021 (HK$13,058.9m) was 60.66 percent of the branch's total advances to banks.

Source: CDB Hong Kong Branch HKMA disclosure statements, extracted in hkma_positions.csv and the OCR pass over the scanned statements (OCR-BACKLOG-RESULTS.md, section 2).

at HK$29.680 million for five, while its mainland row reports impaired nil and overdue nil at all 26 reporting dates from December 2013 to June 2026. Bank of Communications Hong Kong Branch cut its mainland gross book by 77 percent in a single year (2018) and saw Hong Kong impaired advances jump 12.5 times in the first half of 2020, then freeze for five periods. Portfolios do not move this way; single legacy items carried at fixed values, then recognition waves, do. The template does not explain any of it, because explanation is not part of the form. It prints the number, and the number is enough to establish that the aggregate layer above it cannot be reconciled to the branch record by any reader.

7.3 The Venezuela case

The template's sharpest product is the only distress series for a Chinese policy bank loan in the public record. Venezuela's borrower of record is BANDES, a bank, and the lending branch books in Hong Kong, so the position appears in China Development Bank Hong Kong Branch's statements as impaired advances to banks, by country. The borrower-side record describes the protection: an oil escrow machine in which PDVSA sells crude to ChinaOil, ChinaOil pays a BANDES collection account at CDB, debt service is taken at source, a minimum balance of at least 1.3 times the next repayment period is maintained, withdrawals are barred in the 35 days before any repayment date, and CDB may appropriate and set off balances at any time and without notice. The lender-side record values that machine at every reporting date from December 2020 to June 2026: collateral recognised, nil.

Three features of the series carry the argument. First, the allowance was held at exactly 25.00 percent of gross for six consecutive reporting dates while the position was 100 percent overdue over one year at every one of them. Provisioning was formulaic, not reassessed, for two and a half years; the catch-up then arrived in two steps, 41.3 percent at December 2023 and 72.5 percent at June 2024,

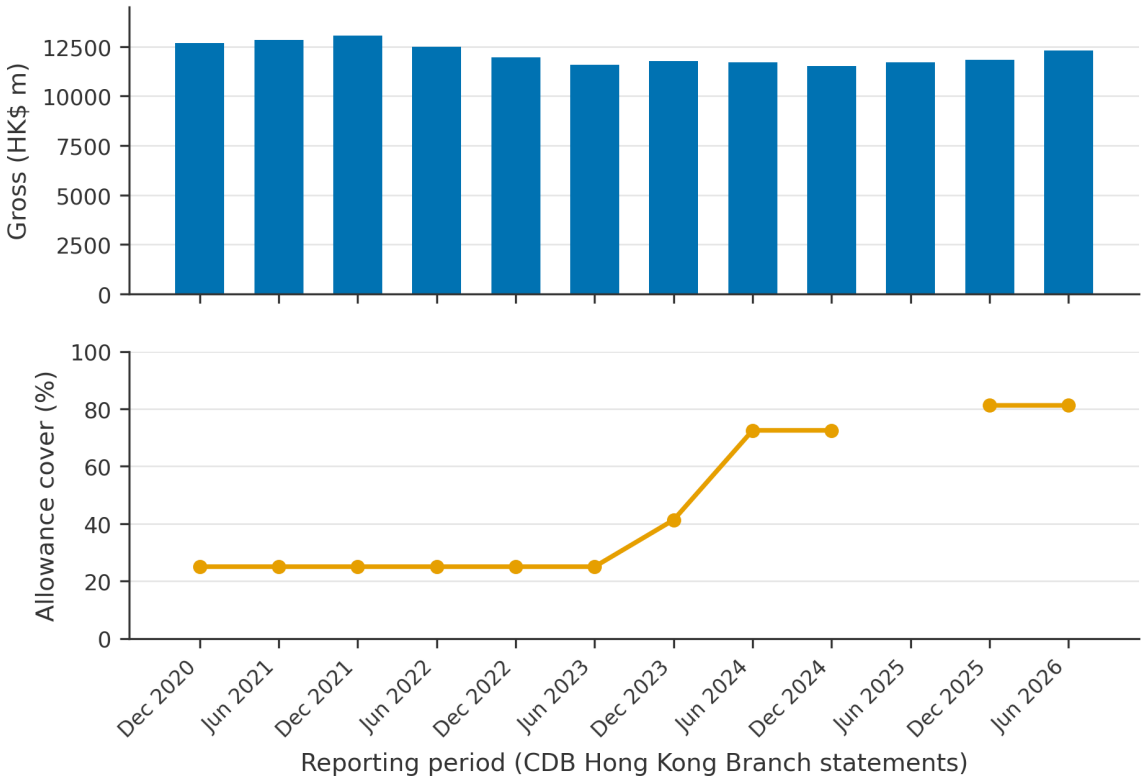


Figure 5: The Venezuela position: gross exposure and allowance cover
Source: CDB Hong Kong Branch disclosure statements. Note: Stacked panels share the reporting period axis. The cover line breaks at June 2025, where the extraction table does not print the allowance. Collateral recognised is nil at all twelve dates and so has no line.

reaching 81.2 percent by December 2025. Second, the gross balance peaked at December 2021 and fell about 12 percent to December 2024 with no repayment, write-off or restructuring recorded in any statement, and has risen in every period since; the statements do not explain the movement. Third, the two records are not contradictory so much as differently honest: an escrow over oil proceeds is a claim on a flow that exists only while the borrower keeps shipping, and a Hong Kong regulator recognises collateral it can value and enforce, of which there was none. Protection, in this case, was worth exactly what the flow was worth, and the flow had stopped.

The branch's customer geography table supplies a footnote that belongs in Section 9: the December 2024 annual statement splits gross customer advances as China HK$5,129.3 million, Hong Kong HK$68,505.2 million, Other HK$6,494.5 million; the June 2025 interim's comparative column for the same date and the same total (HK$80,129.0 million) splits it as China HK$29,060.7 million and Hong Kong HK$51,068.3 million, with the impaired figures identical in both publications. HK$23.9 billion of gross exposure changed geography between publications, consistent with a risk-transfer reallocation, and any series built from these tables inherits the break. The Venezuela rows agree exactly in both.

8 Restructurings and what survives

A restructuring or renegotiation record is held for 35 of the 102 deals, 30 of them Chinese-class, concentrated in the Sri Lanka, Zambia, Chad, Congo, Djibouti and Venezuela clusters. The set is

large enough to test the protections of Section 6 against the event they were written for, and the results split by mechanism type.

The pattern has a single organising variable: whether the protection is a claim on cash outside the borrower's jurisdiction, a claim on cash inside it, a clause, or an asset. Offshore cash claims performed as written, because performance required nothing of the borrower. Onshore cash claims performed exactly as well as the borrower's own monetary authority and audit institutions allowed, which the auditors' reports then recorded: the same instrument produced a performing loan at Hwange, an empty account at Kariba and a central-bank diversion at RGM Airport, three sibling facilities of one lender in one country. Clauses performed worst relative to their drafting. The no-comparability undertaking was written into both Hambantota texts and the Kenyan, Cameroonian, Ecuadorian and Serbian contracts, and in the only systemic default that tested it, Sri Lanka's, comparability was imposed by the structure of the official process rather than conceded by the borrower; the clause is a negotiating position that a systemic default overrides. Asset pledges were never reached: every distressed asset-backed case in the corpus resolved by prepayment, ownership transfer or renegotiation before enforcement, so the corpus contains no observation of what a policy-bank asset pledge is worth in execution, and any claim about port or dam seizure as a recovery strategy is unsupported by this record in either direction.

Two further regularities deserve record. Restructuring is itself a disclosure event: Congo RN1's current rate build is knowable only from the April 2019 restructuring, Chad's three partially knowable loans are its three rescheduled ones, and 30 of the 35 restructuring records attach to Chinese-class deals, so the depth map of Section 4 is partly a distress map. And restructuring can reprice upward: Congo's 2019 rescheduling carried no principal haircut and, on the coded reading, a rate increase on the rescheduled balance, delivered bilaterally a year before the Common Framework existed. Maturity extension as a repricing instrument, executed inside an escrow that was never released, is a different event from the concessional rescheduling the term usually denotes.

9 Anomalies

Anomalies are a first-class layer of this corpus rather than noise around it: in a record built from announcements, annexes, restatements and incompatible publications, the contradictions are often the most informative observations. The archive maintains a formal register (entries A1 to A6, contradictions logged; B1 to B11, signals; D entries, curiosities). This section reports the register with a desk classification of each item into three classes: measurement artefacts (the record disagrees with itself for accounting or editorial reasons), disclosure behaviour (a party chose what to publish, or the choice is visible in the shape of publication), and genuine economic anomalies (the numbers, taken at face value, describe something real and unexplained). The classification is the desk's analysis; the entries are the corpus's.

9.1 Contradictions between a party's own documents (A1 to A6)

A1 (measurement artefact with a disclosure flavour). UKEF's ministerial announcement of the Cairo Monorail guarantee is GBP 1.7 billion; UKEF's own 2020 to 2021 annual report annex lists a combined facility of GBP 936,685,695 for the same asset, and the discrepancy has never been reconciled in any publication held. A2 (reporting error, resolved). Kafue Gorge Lower's financing was attributed in 2019 by Zambia's ambassador to China to "a consortium led by CDB"; the closed record is Eximbank as lead lender with an ICBC tranche, a reminder that official statements about lender

identity are unreliable at exactly the point where public debate uses them most. A3 (our own error, corrected). An earlier corpus note attributed the National Audit Office's concealed-NPL finding to the FY2023 rectification report; the figure (RMB 3.748 billion, two policy banks) is from the FY2024 report, 22 December 2025, page 15. The register keeps the correction because the corpus's citation discipline is part of its method, and because the error is the kind that propagates: a year off in a citation makes the finding unverifiable to any reader who checks. A4 (measurement correction). The first-pass verdict that CCB (Asia) publishes no country table was wrong; the bank does publish one, and the corrected extraction is what makes the B4 anomaly visible. A5 (manifest error). A banked PowerChina settlement announcement (Lin 2026-034, dated 23 July 2026) proves on reading to concern a Basra desalination subcontract of about RMB 8.925 billion, not the Belgrade metro the manifest claimed; document-for-document verification against manifest descriptions is now a standing step. A6 (recovery overturns). Two documents previously graded as having no public route (Kenya's PIDA Hansard record and the Naivasha ICDP gazette notice) were recovered in a later pass; negative verdicts in this archive are provisional by construction.

9.2 Signals in the lender-side record (B1 to B11)

B1 (disclosure behaviour, the paper's central case). CDB Hong Kong's Venezuela allowances held at exactly 25.00 percent for six consecutive reporting dates while the position was 100 percent overdue over one year at every date; cover stepped to 41.3, 72.5 and 81.2 percent in 2023 to 2025. Section 7 sets out the series. B2 (disclosure behaviour). ABC Hong Kong Branch's frozen impaired balances (Hong Kong about HK$88 to 90 million for thirteen periods; mainland about HK$23 million for seven, then HK$795 million, HK$1,679 million, nil), with head office country-risk provision nil throughout: recognition appears as a wave that the branch's own statements do not narrate. B3 (disclosure behaviour). BoCom Hong Kong Branch's mainland book fell 77 percent in 2018 and its Hong Kong impaired balance rose 12.5 times in the first half of 2020, then froze for about five periods. B4 (measurement artefact, unresolved). CCB (Asia)'s "Others" geography: HK$16,079 million (June 2018), HK$44,695 million (June 2019), HK$18,597 million (December 2019), impaired nil at all three dates. A HK$28.6 billion movement into and out of an unnamed residual bucket in eighteen months, with no impairment consequence at any point, is not explained by any document held. B5 (measurement artefact, structural). BOCHK's "Others" bucket grew from HK$622 billion to HK$954 billion in two years, carrying Hong Kong impaired advances up 13 times over 2021 to 2025; the bucket pools ASEAN subsidiaries, so group distress outside Hong Kong and the mainland is unnameable by construction. B6 (disclosure behaviour). ICBC (Asia)'s buffer-table exposures whipsaw by orders of magnitude (Philippines HK$544 thousand to HK$524.5 million to HK$189 thousand; Cambodia HK$63 thousand to HK$2.76 billion to HK$812.1 million), recording portfolio exits that no publication announces. B7 (disclosure behaviour). Sinosure's 2025 claims year (cost up 24.6 percent, profit down 86 percent) is published without a single named claim, in a year when a named claim is the unit its peer EKN publishes. B8 (genuine economic anomaly, the paper's thesis). The 2 percent headline rate: the premium, trapped-cash and currency components of Sections 5 and 6 are not anomalies of individual deals but a systematic gap between the quoted price and the full price, measured wherever the documents survive. B9 (genuine economic anomaly). The Lauca attribution gap: of the USD 4.3 billion Lauca hydropower project in Angola, about USD 220 million of financing is attributed to named lenders (DBSA USD 70 million, Gemcorp USD 150 million) and about USD 4.1 billion is not attributed to anyone in any document held; DBSA's own case study describes a USD 4 billion sovereign loan with BNDES and Deutsche Bank as mandated lead arrangers, and MIGA proposed a USD 750 million guarantee against Breach of Contract that was never issued. The largest single financing in the corpus's region has no lender of record for 95 percent of its value. B10 (disclosure

behaviour). Ghana's Sinohydro facility: the lender of record is the contractor; the receivable sits at CCB; Ghana's total liabilities to China omitted the facility from 2018 to 2021 in debt-sustainability presentations and official debt lists, surfacing first in the December 2024 Eurobond prospectus. B11 (measurement artefact, systematic). Signed against announced amounts: EIB's Bujagali signature is EUR 92,134,679.22 against USD 130 million approved; MIGA's Nachtigal cover issued at EUR 164.5 million against USD 262.5 million approved; IsDB's Senegal TER Phase 2 financing is EUR 140.25 million in total; EACOP's documented closed amount is USD 755 million against a USD 1.0 billion manifest figure; and Mambilla's USD 4.93 billion sits inside Eximbank's sole-lender total although the facility never closed. Announcement-based lending totals inherit an upward bias of unknown size from this class alone, and the corpus's own USD 240.1 billion total carries the Mambilla item flagged rather than removed because the flag, not the adjustment, is what the record supports.

9.3 Contradictions logged, and curiosities

The contradiction log closes every entry it opens. Two entries are signals rather than contradictions: CDB's group NPL ratio of 0.34 percent at 2025 against its Hong Kong branch's Venezuela book is not a contradiction, because concentration inside a group balance sheet is exactly what group ratios conceal; and ICBC Hong Kong Branch's 26 consecutive dates of mainland impaired nil sit against group-level overseas NPL unknowns for the same reason. The curiosities earn their lines because they bound what the archive can know: 124 of the 161 original HKMA grid files proved to be "Page not found" error prints rather than statements, which is why the extraction had to be rebuilt from the files themselves; Eximbank's English site serves a 5-page excerpt as its 2025 annual report; and the Nam Ou cascade's size is stated as about USD 2.4 billion in English sources and about USD 2.8 billion in Chinese sources with no reconciliation in either. None of the three changes any conclusion. All three change how much of the public record a reader should treat as load-bearing.

10 Robustness and falsifiers

This is a forensic record study, not a pre-registered design; the robustness questions are therefore about measurement and selection rather than estimation. Four checks discipline the conclusions.

Selection on documents. The paper's strongest claims (premium levels, clause families, account mechanics) rest on the 23 deals with full or partial contract texts. Section 5's fee result shows the mechanism of the risk: dimensions that exist only in documents are zero in reported-only deals, so a naive read of the matrix would conclude that transparent deals are more expensive and more protected, which is an artefact. The claims that survive are comparative within the documented set (offshore against onshore outcomes; insurance payouts against clause performance) and the claims about missingness itself, which is measured on all 102. The claim that does not survive a selection correction is any statement about levels across the whole corpus: the paper makes none.

The vintage pattern. The 2011 to 2014 escrow peak (52 percent of the band's deals, against 14 and 29 percent in the adjacent bands) could be disclosure lag: later deals have had fewer events to expose their packages. Two observations bound the lag explanation. The peak band's deals are also the richest in recorded dimensions overall (mean 6.9 held of 13, against 6.6 pre-2011 and 5.9 for 2015 to 2019), and distress, the main exposure event, is concentrated in exactly the peak band's deals, which biases its measured protection upward. But the clause families are dated, not inferred: the no-comparability, consent-right and opinion-acceleration forms documented in the corpus are contracts of 2011 to 2014, and the later documented texts in the corpus (the Philippines 2022 railway

credits, Pakistan M5) take materially lighter packages on their face. The desk reads the peak as a real hardening of the template in those years, followed by a period in which two things changed at once: new lending shifted toward borrowers and structures with lighter packages, and the record of the later deals is thinner. The split between those two causes is not identified by this corpus, and the paper does not claim it.

Coding sensitivity. The depth tiers move on single dimensions at the margin: 51 deals carry at least one adjudication override, and the Busanga refresh moved one tier placement on one newly documented position. Re-running the tier rule with the premium dimension dropped entirely (the dimension most exposed to judgement calls) moves no deal between DEEP and the rest, because no DEEP placement depends on a premium coding. Dropping the escrow-location attribution for the 10 deals where location is not stated changes the offshore share of known locations from 11 of 20 to 11 of 30; the paper quotes both forms where it matters.

Falsifiers. The following would change the conclusions, and each is checkable against future publication. (i) If a Chinese policy bank or Sinosure publishes facility-level terms (rate, premium or security) for any corpus project, the China-layer claim is weakened in proportion to the coverage; at present the count is zero across the three institutions' banked publications in the archive. (ii) If any asset pledge in the corpus is documented as enforced, the Section 8 result that pledges are untested falls. (iii) If a borrower-side archive surfaces showing the Venezuela escrow balances actually received by BANDES after 2020, the nil-collateral reading of Section 7 would need revision; at present no such paper exists in the archive. (iv) If contract texts for the 2015 to 2019 band reach the archive at the rate the 2011 to 2014 texts did, and show escrow packages at the peak band's frequency, the template-hardening reading of Section 6 falls and the lag reading stands alone. (v) The census's working claim (no access-to-information statute by itself has ever produced a sovereign loan contract) is stated in the census as falsifiable by a single counterexample; none is recorded in the 34 countries.

11 Limitations

The corpus is the three largest projects per country in a 34-country frame chosen for Chinese and policy-bank lending exposure; it is not a sample of global sovereign lending, and shares computed on it (escrow by lender class, premium ranges) describe this universe, not the population of all policy-bank loans. Committed amounts mix documented close figures with announced figures where close figures do not exist, and the total carries one never-closed facility (Mambilla, USD 4.93 billion) as coded. The premium evidence is six facilities of 22 with documented cover; the range 5.26 to 11.6 percent is a range of observed values, not an estimated distribution, and no average premium is quoted anywhere in the paper for that reason. Rate builds are coded as held or not held, never harmonised, so no pricing-level comparison across deals is attempted. The HKMA extraction rests on OCR of 47 scanned statements; every figure used was cross-checked against the statement's own totals, but OCR risk is not zero and the extraction files are preserved for replication. Documented absence is a strong coding: it asserts that a full or partial text was read and the item is not in it; where texts are partial (10 deals), absence is coded only for the parts held. Finally, the archive's negative verdicts are provisional by construction: two "no public route" verdicts were overturned during the work itself, and three banked documents sit on dead hosts whose contents are unknown rather than zero. Every count in this paper is a count as of the October 2026 archive, and the archive's own log records which of its claims have already been revised once.

12 Conclusion

Read as one record, 102 sovereign loan files describe a lending system whose economics are written in its protections. The interest rate, the number every summary carries, is the component the record holds best and the one that matters least to the deal's real price: on top of it sit insurance premia of 5.26 to 11.6 percent of facility value where they can be seen at all, fees that exist in the data only where a contract text survived, trapped balances of the order of 14 percent of a facility's face, and currency exposure that the borrower carries whole. The protections are the more honest document. A lender that takes an offshore account over oil proceeds, a levy on all imports, a 1.5 times reserve and a consent right over the borrower's other borrowing has stated its assessment of the borrower more precisely than any margin does, and the restructuring record shows the statement was accurate: offshore cash performed, onshore cash performed as well as the borrower's institutions allowed, clauses yielded to the first systemic default, insurance paid, and no asset was ever seized.

The disclosure layer is asymmetric by design on both sides. Borrower states publish contracts through ratification channels, audit institutions and distress, in patterns that track their institutions rather than their debts; the depth audit shows the resulting record is richest exactly where a deal has already failed. The Chinese lending institutions publish aggregates that are accurate and uninformative in the same document, and facility-level terms nowhere; the only window into the system's distressed book is a host regulator's template in Hong Kong, which values a decade of oil-escrow architecture at nil collateral and records provisioning that sat at a round 25 percent for six reporting dates on a position overdue in full. The 99 of 122 counterparties that publish nothing at facility level are not a gap in this archive. They are the finding the archive was built to measure.

For practice, three implications follow. Debt sustainability work that prices these loans at their headline rates understates their cost by the premium, the trapped cash and the fees, and overstates the borrower's liquidity by the balances it cannot touch; both errors are largest for the escrowed deals, which are the deals most likely to be distressed. Restructuring design should treat offshore cash claims as senior in fact whatever the legal ranking says, because the record shows them collecting through treatments in which unsecured claims were extended; the comparability debate is, in this corpus, a debate about whether that fact is acknowledged in the term sheet or discovered in the escrow statements. And transparency reform aimed at statutes alone will keep failing on the census's evidence: what produced contracts in 34 countries was ratification pipelines with publication duties, audit institutions with contract access, and the disclosure that distress forces. The next pass of this programme extends the census to a second wave of countries and re-runs the depth audit on the archive as it grows; the falsifiers in Section 10 state what would change the conclusions, and the archive's correction log is published with the data so that its own revisions are part of the record.

Production note: This paper was produced by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data. The underlying corpus (dossiers, coding matrix, counterparty archive) is maintained by Kilambi Research; document-level sources are listed in Appendix C.

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Table 6: Protection types in distress: outcomes recorded in the corpus

Protection	Recorded outcome
Offshore escrow / collection	Survived and kept collecting. Kenya's 2025 restructuring (currency conversion, extension to 2040) left the trap intact, KES 44.63bn still locked in Oct 2026. Congo's escrow was not released in 2019; the rescheduling repriced the rate upward on the rescheduled 67 percent of the balance, an NPV gain to the creditor of about 23 percent.
Onshore escrow	Held where the borrower's institutions respected it (Hwange: USD 48.46m in escrow at Dec 2024, the only one of eight on-lent Eximbank loans with zero arrears); failed where they reached it (Kariba South: balance USD 0.00 against USD 36.94m arrears after the borrower netted proceeds away; RGM Airport: USD 10m diverted by the central bank in 2019, restored only after disbursement freezes on three projects).
No-comparability clauses	Did not hold. Sri Lanka's Eximbank texts carry the clause in both contracts; the 2024 treatment was comparable to the official creditor committee's, arrears capitalised, maturities extended to 2043.
Insurance	Paid. Sinosure indemnities of about USD 200m on the Ethiopian tranches; USD 57.23m at Kafue Gorge Lower, paid to the contractor Sinohydro, which had filed the claim. The only protections in the distressed set with documented cash payouts.
Asset pledges	Not tested. No asset pledge in the corpus is documented as enforced. Nacala's lenders exited by sponsor prepayment at par; Tonkolili was resolved by ownership transfer inside the sponsor group; the Sicomines mining security has been renegotiated (2017, 2024), not enforced.
Reserves in local currency	Eroded without default. Ghana's DSRA, frozen at GHS 370,586,909 from 2017 to 2022, secured a dollar debt; the trap held and the cover shrank each year the cedi fell.
Security renegotiated upward	Twice. Pakistan's PERF (PKR 50bn escrow, Oct 2022, extracted by the insurer mid-life); Beineu-Bozoi-Shymkent's onshore pledge, added after a milestone breach was cured by a USD 400m prepayment.

Notes: Outcomes are as documented in the dossiers and the collateral pass; "documented" follows the coding of Section 4.2.

Source: COLLATERAL-PRACTICES.md, section 4.3.

A Appendix A: The depth audit in detail

The 13 coded dimensions, with held counts by coding (P = held from a document, R = held as reported only):

Dimension	P	R	Held (of 102)
Terms read completed	102	0	102
Tenor and grace	62	25	87
Security package	59	21	80
Official terms document	74	0	74
Interest rate build	56	18	74
Confidentiality / no-comparability position	37	9	46
Restructuring record	33	2	35
Fees	34	0	34
Escrow or revenue capture	24	6	30
Lender-side document	30	0	30
Governing law and arbitration	14	1	15
Full contract text	13	0	13
Insurance premium	3	3	6

Notes: Counts are the refreshed audit (October 2026). The terms read is a desk product and is complete for all 102 dossiers by construction.
Source: DATA-DEPTH-MATRIX.csv.

The mechanical tier rule: DEEP requires an official terms document or full contract, the rate build held, and the security package documented. MEDIUM requires a document base (official terms document or full contract) with at least five dimensions held, or at least seven dimensions without a document base. THIN is at least three dimensions and not DEEP or MEDIUM. MINIMAL is two or fewer. Refreshed tiers: DEEP 48 deals (USD 111.8bn), MEDIUM 23 (USD 54.9bn), THIN 24 (USD 59.8bn), MINIMAL 7 (USD 13.6bn).

Adjudication corrections that define the counts (from the audit's review file): interest rate swap premia (Kenya SGR) and ADB maturity premia are not insurance premia; a premium quoted comparatively inside another deal's dossier (the Kribi premium inside Gabon's dossier) belongs to that other deal; concession and BOT terms are not loan terms (Kafue's 4.7455 percent DRS average and Nachtigal's modelled rates are not facility terms and are excluded); gap lists in dossiers enumerate what is not known and were excluded as evidence throughout; a lender-side document must be issued by the lender, a counterparty, or a regulator disclosing that lender.

B Appendix B: The 102 deals

Table 7: The deal corpus: project, country, lenders, committed amount, depth tier and dimensions held

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
1	Mombasa-Nairobi Standard Gauge Railway (Phase 1)	Kenya	China Eximbank (concessional + commercial tranches)	3,233	DEEP	12

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
2	Nairobi-Naivasha Standard Gauge Railway (Phase 2A)	Kenya	China Eximbank	1,482	DEEP	11
3	Olkaria I and IV Geothermal Expansion (280 MW) under Kenya Electricity Expansion Project	Kenya	World Bank (IDA); Japan International Cooperation Agency (JICA); Agence Francaise de Develop...	957	DEEP	7
4	Kafue Gorge Lower Hydropower Plant (750 MW)	Zambia	China Eximbank and Industrial and Commercial Bank of China (ICBC) syndicate	1,531	MEDIUM	6
5	Lusaka-Ndola Dual Carriageway (321 km)	Zambia	China Eximbank (Jiangxi Branch); Jiangxi Bank	1,246	DEEP	6
6	Maamba Collieries Integrated Coal Mine and 300 MW Power Plant (Phase I)	Zambia	Bank of China; ICBC; Standard Chartered; Industrial Development Corporation of South Africa;...	590	DEEP	6
7	Caculo Cabaca Dam and Transmission Line (2,172 MW)	Angola	ICBC; China Eximbank; Bank of China; China Construction Bank; China Minsheng Bank; Ping An B...	4,131	THIN	4
8	Lauca Dam (2,070 MW)	Angola	Sovereign loan arranged by BNDES and Deutsche Bank (mandated lead arrangers, per DBSA Sustai...	4,300	MINIMAL	2
9	New Luanda International Airport (Dr Antonio Agostinho Neto International Airport)	Angola	ICBC and China CITIC Bank (syndicated buyer's credit); China Development Bank (separate loan...	1,289	THIN	3
10	Nacala Logistics Corridor Railway and Port (Moatize-Nacala)	Mozambique	Japan Bank for International Cooperation (JBIC); NEXI-insured commercial syndicate (SMBC, MU...	2,730	MEDIUM	6
11	Maputo-Katembe Bridge and Link Roads	Mozambique	China Eximbank	682	THIN	6

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
12	Central Termica de Temane (CTT) 450 MW Gas-to-Power Plant	Mozambique	International Finance Corporation (IFC) with B-loan participants FMO and Emerging Africa Inf...	495	MEDIUM	6
13	Addis Ababa-Djibouti Railway (Ethiopian section)	Ethiopia	China Eximbank	2,491	THIN	6
14	National Telecommunications Network Infrastructure Expansion Project, Phases 1-3 (ZTE vendor finance)	Ethiopia	ZTE Corporation supplier's credit, backed by a China Development Bank (CDB) export seller's ...	1,531	THIN	5
15	Awash-Kombolcha-Hara Gebeya (Awash-Weldiya) Railway	Ethiopia	Credit Suisse-led syndicate; Turk Eximbank (USD 300m); European export credit agencies (Denm...	865	THIN	4
16	Karuma Hydropower Project (600 MW)	Uganda	China Eximbank	1,400	DEEP	8
17	Bujagali Hydropower Project (250 MW)	Uganda	IFC; African Development Bank; European Investment Bank; Proparco; FMO; DEG/KfW; Absa; Stand...	682	DEEP	8
18	Isimba Hydropower Project (183 MW)	Uganda	China Eximbank	482	DEEP	7
19	Standard Gauge Railway Dar es Salaam to Mwanza Phase 1 (Lots 1-5)	Tanzania	Standard Chartered (arranger); EKN/SEK (Sweden), EKF (Denmark), SACE (Italy), KUKE (Poland),...	3,790	THIN	5
20	Tanzania-Burundi-DRC Standard Gauge Railway (Tabora-Kigoma and Uvinza-Musongati)	Tanzania	African Development Bank (global coordinator); Deutsche Bank and Societe Generale (co-mandat...	2,150	THIN	4
21	East African Crude Oil Pipeline (EACOP) - Tanzania section	Tanzania	Syndicated loan: Stanbic Bank, KCB Bank, Standard Bank, Afreximbank, Islamic Corporation for...	1,000	MINIMAL	2

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
22	Sicomines Resources-for-Infrastructure Convention (Sino-Congolese programme)	DR Congo	China Eximbank and Chinese consortium (Sinohydro, China Railway Group) via Sicomines JV with...	7,000	DEEP	6
23	Busanga Hydropower Plant (240 MW)	DR Congo	Sinohydro / PowerChina Resources and China Railway Resources Group via Sino-Congolaise Hydro...	660	THIN	3
24	Zongo II Hydropower Dam and 220 kV Transmission Line (150 MW)	DR Congo	China Eximbank	360	DEEP	7
25	China Development Bank Master Facility - Western Corridor Gas / Atuabo and other infrastructure	Ghana	China Development Bank	3,000	DEEP	7
26	Sinohydro Bauxite-for-Infrastructure Master Project Support Agreement	Ghana	Sinohydro Corporation (deferred-payment EPC; refinanced via China Construction Bank accounts...	2,000	DEEP	9
27	Tema Port Expansion (Meridian Port Services)	Ghana	IFC (coordinator, USD 195m own account); Bank of China, ICBC, Standard Bank and FMO (USD 472...	667	MEDIUM	5
28	Kribi Deep Sea Port Phases 1 and 2	Cameroon	China Eximbank	1,100	DEEP	10
29	Nachtigal Hydropower Plant (420 MW)	Cameroon	IFC-coordinated syndicate: IFC, AfDB, AFD/Proparco, EIB, DEG, FMO, CDC Group, Africa Finance...	1,010	MEDIUM	6
30	Memve'ele Hydropower Plant (211 MW)	Cameroon	China Eximbank	683	DEEP	10
31	Dakar Regional Express Train (TER) Phases 1 and 2	Senegal	Islamic Development Bank, African Development Bank, AFD, French Treasury (direct loans), BOA...	1,300	MEDIUM	6
32	Dakar Bus Rapid Transit (Electric BRT)	Senegal	World Bank IDA (USD 300m + USD 70m additional financing), European Investment Bank, Green Cl...	614	MEDIUM	6

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
33	Cap des Biches Combined-Cycle Power Plant - West African Energy (300 MW)	Senegal	Africa Finance Corporation, Coris Bank International, Afreximbank and other institutions	349	MINIMAL	2
34	Mambilla Hydropower Project (3050 MW in Taraba)	Nigeria	China Eximbank (proposed)	4,930	MINIMAL	1
35	Ajaokuta-Kaduna-Kano (AKK) Gas Pipeline	Nigeria	Bank of China; Sinosure (China Export and Credit Insurance Corporation)	2,600	THIN	5
36	Lagos-Ibadan Standard Gauge Railway	Nigeria	China Eximbank	1,267	DEEP	6
37	Pointe-Noire-Brazzaville Road (RN1) - Dolisie to Brazzaville Section	Republic of Congo	China Eximbank	1,000	DEEP	7
38	Power Transmission Network Associated with the Imboulou Hydropower Plant	Republic of Congo	CMEC (supplier's credit); China Eximbank (reported seller's credit behind CMEC)	552	DEEP	10
39	Imboulou Hydropower Plant (120 MW)	Republic of Congo	CMEC (supplier's credit financed by China Eximbank export seller's credit)	238	THIN	4
40	Glencore Oil-Backed Prepayment Facility (2014)	Chad	Glencore; syndicate of banks	1,450	DEEP	8
41	N'Djamena (Djermaya) Refinery and Pipeline Project	Chad	China Eximbank; CNPC International (shareholder loan)	330	DEEP	6
42	N'Djamena-Djermaya 66kV Transmission Line and 90kV N'Djamena Power Transmission Loop and Distribution Network	Chad	China Eximbank	130	MINIMAL	1
43	Souapiti Hydroelectric Power Plant (450 MW)	Guinea	China Eximbank	1,175	DEEP	7
44	Amaria Hydroelectric Power Plant (300 MW)	Guinea	China Development Bank (CDB)	830	THIN	5
45	Reconstruction of the RN1 Coyah-Mamou-Dabola Road	Guinea	ICBC; China Eximbank (syndicated)	375	DEEP	6
46	Tonkolili Iron Ore Project - Rail and Pepel Port Infrastructure	Sierra Leone	China Development Bank (CDB)	1,200	THIN	4
47	Western Area Power Generation Project (Freetown gas-to-power)	Sierra Leone	US International Development Finance Corporation (DFC); ECOWAS Bank for Investment and Devel...	332	MEDIUM	7

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
48	Bumbuna Hydroelectric Project Phase I (50 MW)	Sierra Leone	African Development Bank (ADF/NTF); Government of Italy; IDA (World Bank); OPEC Fund (OFID);...	327	MEDIUM	5
49	Hwange Thermal Power Station Expansion - Units 7 and 8 (600 MW)	Zimbabwe	China Eximbank	998	DEEP	7
50	Kariba South Hydroelectric Power Station Extension (300 MW)	Zimbabwe	China Eximbank	319	DEEP	7
51	Robert Gabriel Mugabe International Airport Expansion	Zimbabwe	China Eximbank	153	DEEP	7
52	Addis Ababa-Djibouti Railway - Ali Sabieh to Nagad Section (Djibouti share, 100 km)	Djibouti	China Eximbank	492	MEDIUM	6
53	Doraleh Multipurpose Port Phase 1 and Damerjog Livestock Export Terminal	Djibouti	China Eximbank; ICBC	394	MEDIUM	5
54	Ethiopia-Djibouti Water Pipeline Project Phase 1	Djibouti	China Eximbank	322	MEDIUM	6
55	El Dabaa Nuclear Power Plant (4 x VVER-1200, 4,800 MW)	Egypt	Russian state	25,000	DEEP	6
56	New Administrative Capital - Central Business District	Egypt	ICBC-led Chinese bank syndicate (Sinosure-covered)	3,000	THIN	4
57	Cairo Monorail - New Administrative Capital and 6th of October Lines	Egypt	JP Morgan (facility agent / lead arranger) and other financial institutions; UK Export Finan...	2,200	THIN	4
58	Extension of Southern Expressway Project (Matara to Hambantota; Sections 1-4)	Sri Lanka	Export-Import Bank of China (China Eximbank)	1,711	MEDIUM	7
59	Puttalam / Norochcholai Coal Power Project (Lakvijaya; Phases I-II)	Sri Lanka	Export-Import Bank of China (China Eximbank)	1,344	MEDIUM	6
60	Hambantota Port Development Project (Phases I-II plus bunkering facility)	Sri Lanka	Export-Import Bank of China (China Eximbank)	1,322	DEEP	12
61	Sukkur-Multan Motorway (M-5; Sukkur-Multan Section of Karachi-Lahore Motorway)	Pakistan	Export-Import Bank of China (China Eximbank)	2,890	DEEP	8

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
62	China Power Hub Generation Company (CPHGC) Hub Coal Power Project (2x660 MW plus coal jetty)	Pakistan	China Development Bank (lead); China Eximbank; ICBC; China Construction Bank; Bank of Commun...	1,496	DEEP	7
63	Sahiwal Coal Power Plant (2x660 MW)	Pakistan	Industrial and Commercial Bank of China (ICBC)	1,440	DEEP	7
64	Rooppur Nuclear Power Plant (2x1200 MW VVER)	Bangladesh	Russian Federation state export credit (via Rosatom / Atomstroyexport as contractor)	11,385	DEEP	8
65	Matarbari Ultra Super Critical Coal-Fired Power Project (Phase I; 2x600 MW plus coal port)	Bangladesh	Japan International Cooperation Agency (JICA)	5,500	DEEP	8
66	Padma Bridge Rail Link Project	Bangladesh	Export-Import Bank of China (China Eximbank)	2,668	MEDIUM	5
67	China-Laos (Boten-Vientiane) Railway Project	Laos	Export-Import Bank of China (China Eximbank)	4,020	THIN	4
68	Xayaburi Hydropower Project (1285 MW; Mekong mainstream)	Laos	Thai commercial bank consortium (Bangkok Bank; Krungthai; Siam Commercial; Kasikornbank; Ban...	3,800	DEEP	8
69	Nam Ou River Cascade Hydropower Project (Phases 1-2; seven dams; 1272 MW)	Laos	China Development Bank (CDB); China Eximbank; China Construction Bank (Phase 2 syndicate)	1,960	THIN	5
70	Phnom Penh-Sihanoukville Expressway	Cambodia	China Development Bank (lead arranger); China Eximbank (syndicated loan to PPSHV Expressway ...	1,560	MINIMAL	1
71	Phnom Penh-Bavet Expressway	Cambodia	China Road and Bridge Corporation (CRBC) BOT concession; commercial loans (lenders not named...	1,350	MINIMAL	2

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
72	Lower Sesan 2 Hydropower Plant (400 MW)	Cambodia	China Development Bank (Hong Kong Branch; arranger); Bank of China (Phnom Penh Branch); Chin...	781	DEEP	7
73	Jakarta-Bandung High-Speed Railway (Whoosh; KCIC)	Indonesia	China Development Bank (CDB)	4,500	MEDIUM	7
74	Central Java Power Project (Batang; 2x950-1000 MW)	Indonesia	Japan Bank for International Cooperation (JBIC; lead) with MUFG; Mizuho; SMBC; Sumitomo Mits...	3,400	MEDIUM	5
75	Tanjung Jati B Power Plant Expansion (Units 5 and 6; 2x1000 MW)	Indonesia	Japan Bank for International Cooperation (JBIC); MUFG; Mizuho; SMBC; Sumitomo Mitsui Trust; ...	3,355	THIN	4
76	South Commuter Railway Project (North-South Commuter Railway - South)	Philippines	Asian Development Bank (ADB); Japan International Cooperation Agency (JICA) cofinancing roll...	4,300	DEEP	8
77	Malolos-Clark Railway Project (North-South Commuter Railway - North)	Philippines	Asian Development Bank (ADB); Japan International Cooperation Agency (JICA)	2,750	DEEP	8
78	Bataan-Cavite Interlink Bridge	Philippines	Asian Development Bank (ADB); Asian Infrastructure Investment Bank (AIIB)	3,240	DEEP	7
79	Upper Trishuli-1 Hydropower Project (216 MW)	Nepal	IFC; Asian Development Bank (ADB); Asian Infrastructure Investment Bank (AIIB); Export-Impor...	453	MEDIUM	5
80	Tanahu Hydropower Project (140 MW, with storage)	Nepal	Multilateral/DFI	419	DEEP	7
81	Pokhara Regional International Airport	Nepal	Export-Import Bank of China (China Eximbank); China Ministry of Commerce (MOFCOM)	213	DEEP	7
82	Greater Male Connectivity Project - Thilamale Bridge	Maldives	Export-Import Bank of India (India Eximbank) line of credit; Government of India grant	500	THIN	4

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
83	Velana International Airport Upgrade and Expansion Project	Maldives	Export-Import Bank of China (China Eximbank)	374	DEEP	6
84	Sinamale Bridge (China-Maldives Friendship Bridge)	Maldives	Export-Import Bank of China (China Eximbank); China Ministry of Commerce (MOFCOM) grant	222	MEDIUM	5
85	China Development Bank oil-backed facility - June 2011 (USD 2.052 billion facility)	Ecuador	China Development Bank (CDB)	2,052	DEEP	10
86	China Development Bank oil-backed facility - December 2012 (USD 2 billion facility)	Ecuador	China Development Bank (CDB)	2,000	DEEP	7
87	Coca Codo Sinclair Hydropower Project (1500 MW)	Ecuador	Export-Import Bank of China (China Eximbank)	1,680	DEEP	10
88	Long-Term / Grand Volume Financing Facility (Fondo de Gran Volumen a Largo Plazo)	Venezuela	China Development Bank (CDB)	20,300	MEDIUM	8
89	Joint Chinese-Venezuelan Fund - Tranche C (USD 5 billion)	Venezuela	China Development Bank (CDB) via BANDES	5,000	MEDIUM	7
90	Joint Chinese-Venezuelan Fund Tranche B Renewal, Second Rollover (2015 package, incl. Sep 2015 PDVSA oil-field leg)	Venezuela	China Development Bank (CDB)	5,000	MEDIUM	7
91	Bar-Boljare Motorway - Priority Section Smokovac-Matesevo (Phase 1)	Montenegro	Export-Import Bank of China (China Eximbank)	944	DEEP	8
92	Bar-Boljare Motorway - Section 2 Matesevo-Andrijevica	Montenegro	European Bank for Reconstruction and Development (EBRD); European Union (EU) grant	350	THIN	3
93	Gvozd Wind Farm (54.6 MW, Phase 1)	Montenegro	Multilateral/DFI	90	MEDIUM	5
94	Belgrade-Nis Railway Modernisation (Corridor X, Belgrade-Nis section)	Serbia	European Investment Bank (EIB); European Bank for Reconstruction and Development (EBRD)	1,650	DEEP	10
95	Hungarian-Serbian Railway - Novi Sad-Subotica-State Border (Kelebija) Section	Serbia	Export-Import Bank of China (China Eximbank)	988	DEEP	9
96	Kostolac B Power Plant Package Project (Phase 1 B1/B2 rehabilitation + Phase 2 new B3 unit and Drmno mine expansion)	Serbia	Export-Import Bank of China (China Eximbank)	901	DEEP	9

#	Project	Country	Lender(s), as coded	USD m	Tier	Dims (of 13)
97	Central Asia-China Gas Pipeline - Kazakh Section (Lines A and B + Line C)	Kazakhstan	China Development Bank (CDB); Bank of China (BOC)	12,200	THIN	4
98	Beineu-Bozoi-Shymkent Gas Pipeline (Beineu-Shymkent Gas Pipeline)	Kazakhstan	China Development Bank (CDB); Bank of China (BOC)	2,500	THIN	6
99	Atyrau Refinery Modernisation (Aromatic Hydrocarbons Complex + Advanced/Deeper Oil Conversion Complex)	Kazakhstan	Export-Import Bank of China (China Eximbank); Japan Bank for International Cooperation (JBIC)	2,312	THIN	4
100	Central Asia-China Gas Pipeline - Uzbek Sections (Lines A and B + Line C)	Uzbekistan	China Development Bank (CDB); Bank of China (BOC)	4,900	THIN	3
101	Ustyurt Gas Chemical Complex at Surgil Gas Field (Surgil / Uz-Kor Gas Chemical)	Uzbekistan	Export-Import Bank of Korea (KEXIM); Korea Trade Insurance Corporation (KSure); Asian Develo...	2,540	DEEP	7
102	Uzbekistan GTL Plant (Oltin Yo'l GTL, Shurtan, Kashkadarya)	Uzbekistan	China Development Bank (CDB); Export-Import Bank of Korea (KEXIM); Korea Trade Insurance Cor...	2,300	DEEP	6

Notes: Lender strings are as coded in PROJECT-UNIVERSE.csv, truncated where the syndicate list is long. Committed amounts are documented close figures where held and announced figures otherwise; Mambilla (row 34) never closed and is included as coded. Dimensions held counts P and R codings across the 13 dimensions of Appendix A.

C Appendix C: Sources and series definitions

Corpus files. The paper's quantitative content derives from the Kilambi Research disclosure programme archive: the disclosure census (34 countries, recipient-publication grading protocol); the deal universe file (102 projects: country, region, sector string, lender string, committed amount, signing year); the data-depth matrix (102 rows, 13 coded dimensions, P/R codings, tier assignments, adjudication overrides preserved in a companion file); the collateral coding pass (27 mechanism categories, per-deal classifications, case files); the anomaly register and contradiction log; the counterparty evidence refresh (per-party facility-level publication verdicts for 122 parties); the China-layer reads (Eximbank, CDB group, Sinosure annual-report series); and the HKMA extraction (340 position rows across six institutions' disclosure statements, with a seventh grid institution, Bank of China (Hong Kong), read from its banked statements in the same pass; an OCR pass over 47 scanned statements, figures cross-checked against statement totals).

Series definitions. *Committed amount*: documented close figure where held, announced figure

otherwise; the Mambilla facility is included as coded and flagged. *Lender class*: coded from the dossier lender string; a deal is Chinese policy-bank if China Eximbank or CDB is a lender of record; multilateral/DFI if a multilateral or development finance institution leads; Russian state for the Rosatom credits; other bilateral/commercial otherwise. Joint deals follow the lead lender of record. *Region*: as coded in the universe file. *Sector group*: desk construction by keyword rule over the universe file's 41 unstandardised sector strings (energy: power, hydro, coal, gas, oil, solar, wind, geothermal, transmission; transport: rail, road, port, airport, bridge, metro, BRT). *Vintage band*: signing year from the universe file; 3 deals carry no signing year and are excluded from vintage statistics. *Escrow held*: P or R coding on the escrow dimension; location (offshore, onshore, mixed, unstated) as coded in the matrix. *Cover (%)*: allowances divided by gross impaired advances, as printed in the branch statements. *HKMA position row*: one institution, one period, one geography or category line in the extraction file.

Census grading protocol (summary). HIGH: full contract texts obtainable for a substantial share of major foreign infrastructure loans through a domestic channel that exists now. MEDIUM: channels have produced full texts or clause-level disclosure for flagship deals, episodically. LOW: no realistic route to contract texts; terms surface only in aggregates, restructurings or leaks. Grades assess recipient publication only; lender publication is coded separately in the counterparty pass.